

SQL Project: Retail Sales Analysis

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🔑 Database Setup

```
CREATE DATABASE retail_;
USE retail_;
CREATE TABLE retail_sales(
  transactions_id INT PRIMARY KEY,
  sale_date DATE,
  sale_time TIME,
  customer_id INT,
  gender VARCHAR(10),
  age INT,
  category VARCHAR(100),
  quantity INT,
  price_per_unit FLOAT,
  cogs FLOAT,
  total_sale FLOAT
);
```

🔍 Basic Data Checks

```
-- Record Count
SELECT COUNT(*) FROM retail_sales;
```

Output:

COUNT(*)
1987

-- Unique Customers

```
SELECT COUNT(DISTINCT customer_id) AS unique_cust FROM retail_sales;
```

Output:

unique_cust
155

-- Unique Categories

```
SELECT COUNT(DISTINCT category) AS unique_category FROM retail_sales;
```

Output:

unique_category
3

-- Null Check

```
SELECT * FROM retail_sales
```

```
WHERE sale_date IS NULL OR sale_time IS NULL OR customer_id IS NULL OR gender IS  
NULL OR
```

```
    age IS NULL OR category IS NULL OR quantity IS NULL OR price_per_unit IS NULL OR  
    cogs IS NULL;
```

-- Deleting Nulls

```
SET SQL_SAFE_UPDATES = 1;
```

```
DELETE FROM retail_sales
```

```
WHERE sale_date IS NULL OR sale_time IS NULL OR customer_id IS NULL OR gender IS  
NULL OR
```

```
    age IS NULL OR category IS NULL OR quantity IS NULL OR price_per_unit IS NULL OR  
    cogs IS NULL;
```

Data Analysis Queries

Q1: Sales made on '2022-11-05'

```
SELECT * FROM retail_sales
```

```
WHERE sale_date='2022-11-05';
```

Output:

transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantity	price_per_unit	cogs	total_sale
1137	2022-11-05	22:34:00	104	Male	46	Beauty	2	500	145	1000
943	2022-11-05	19:29:00	90	Female	57	Clothing	4	300	318	1200
856	2022-11-05	17:43:00	102	Male	54	Electronics	4	30	9.3	120
240	2022-11-05	11:49:00	95	Female	23	Beauty	1	300	123	300
214	2022-11-05	16:31:00	53	Male	20	Beauty	2	30	8.1	60
180	2022-11-05	10:47:00	117	Male	41	Clothing	3	300	129	900
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Q2: Clothing category, quantity > 4 in November 2022

```
SELECT * FROM retail_sales
WHERE category ='clothing' AND DATE_FORMAT(sale_date, '%Y-%m')='2022-11'
AND quantity >= 4;
```

Output:

transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantity	price_per_unit	cogs	total_sale
64	2022-11-15	06:34:00	7	Male	49	Clothing	4	25	8.5	100
146	2022-11-10	22:01:00	74	Male	38	Clothing	4	50	49	200
159	2022-11-10	21:30:00	42	Male	26	Clothing	4	50	23.5	200
284	2022-11-12	09:17:00	129	Male	43	Clothing	4	50	20.5	200
547	2022-11-14	07:36:00	3	Male	63	Clothing	4	500	250	2000
699	2022-11-21	22:21:00	129	Female	37	Clothing	4	30	16.2	120
735	2022-11-26	21:38:00	153	Female	64	Clothing	4	500	515	2000

Q3: Total sales by category

```
SELECT category, SUM(total_sale) as net_sale, COUNT(*) AS total_orders
FROM retail_sales
GROUP BY category;
```

Output:

category	net_sale	total_orders
Beauty	286790	611
Clothing	309995	698
Electronics	311445	678

Q4: Average age of customers in 'Beauty' category

```
SELECT ROUND(AVG(age), 2) as avg_age  
FROM retail_sales  
WHERE category='Beauty';
```

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
avg_age			
40.42			

Q5: Transactions with total_sale > 1000

```
SELECT * FROM retail_sales  
WHERE total_sale > 1000;
```

Output:

Result Grid				Filter Rows:		Export:		Wrap Cell Content:			
	transactions_id	sale_date	sale_time	customer_id	gender	age	category	quantiy	price_per_unit	cogs	total_sale
▶	522	2022-07-09	11:00:00	52	Male	46	Beauty	3	500	145	1500
	559	2022-12-12	10:48:00	5	Female	40	Clothing	4	300	84	1200
	1522	2022-11-14	08:35:00	48	Male	46	Beauty	3	500	235	1500
	1559	2022-08-20	07:40:00	49	Female	40	Clothing	4	300	144	1200
	421	2022-04-08	08:43:00	66	Female	37	Clothing	3	500	235	1500
	1421	2022-01-17	07:07:00	59	Female	37	Clothing	3	500	185	1500
	484	2022-03-13	07:52:00	135	Female	19	Clothing	4	300	75	1200
	1484	2022-11-23	09:29:00	22	Female	19	Clothing	4	300	147	1200
	15	2022-07-01	11:50:00	75	Female	42	Electronics	4	500	210	2000
	743	2022-08-07	07:54:00	55	Female	34	Beauty	4	500	260	2000
	1015	2022-03-09	11:53:00	94	Female	42	Electronics	4	500	200	2000
	1743	2022-10-26	09:37:00	47	Female	34	Beauty	4	500	250	2000
	742	2022-03-19	06:08:00	37	Female	38	Electronics	4	500	195	2000
	1742	2022-11-22	08:25:00	18	Female	38	Electronics	4	500	220	2000
	420	2022-01-02	10:53:00	28	Female	22	Clothing	4	500	200	2000
	1420	2022-04-15	07:01:00	138	Female	22	Clothing	4	500	205	2000
	592	2022-12-26	09:15:00	77	Female	46	Beauty	4	500	275	2000
	1592	2022-03-16	09:08:00	81	Female	46	Beauty	4	500	155	2000
	720	2022-04-08	08:50:00	116	Female	56	Beauty	3	500	235	1500

Q6: Total transactions by gender in each category

```
SELECT category, gender, COUNT(*) AS total_trans  
FROM retail_sales  
GROUP BY category, gender  
ORDER BY category;
```

Output:

Result Grid



Filter Rows:

Export:

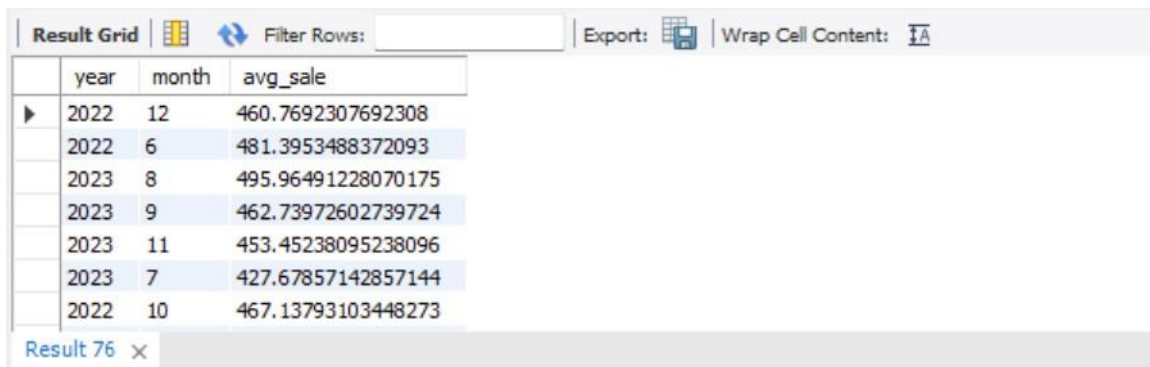

Wrap Cell Content:


	category	gender	total_trans
▶	Beauty	Female	330
	Beauty	Male	281
	Clothing	Female	347
	Clothing	Male	351
	Electronics	Female	335
	Electronics	Male	343

Q7: Average sale for each month and best-selling month

```
SELECT
  EXTRACT(YEAR FROM sale_date) AS year,
  EXTRACT(MONTH FROM sale_date) AS month,
  AVG(total_sale) AS avg_sale
FROM retail_sales
GROUP BY year, month;
```

Output:



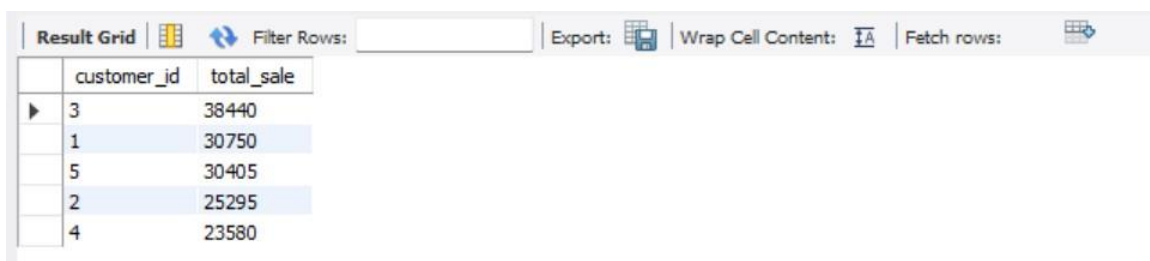
The screenshot shows a database query result grid with columns: year, month, and avg_sale. The data is grouped by year and month. The results are as follows:

year	month	avg_sale
2022	12	460.7692307692308
2022	6	481.3953488372093
2023	8	495.96491228070175
2023	9	462.73972602739724
2023	11	453.45238095238096
2023	7	427.67857142857144
2022	10	467.13793103448273

Q8: Top 5 customers by total sales

```
SELECT
  customer_id, SUM(total_sale) AS total_sale
FROM retail_sales
GROUP BY customer_id
ORDER BY total_sale DESC
LIMIT 5;
```

Output:



The screenshot shows a database query result grid with columns: customer_id and total_sale. The results are as follows:

customer_id	total_sale
3	38440
1	30750
5	30405
2	25295
4	23580

Q9: Unique customers per category

```
SELECT category, COUNT(DISTINCT customer_id) AS uniq_cust
FROM retail_sales
GROUP BY category;
```

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
category	uniq_cust		
Beauty	141		
Clothing	149		
Electronics	144		

Q10: Shifts and number of orders

```

SELECT CASE
  WHEN HOUR(sale_time) < 12 THEN 'Morning'
  WHEN HOUR(sale_time) BETWEEN 12 AND 17 THEN 'Afternoon'
  ELSE 'Evening'
END AS Shift,
COUNT(*) AS num_of_orders
FROM retail_sales
GROUP BY Shift;

```

Output:

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Shift	num_of_orders		
evening	1062		
morning	548		
afternoon	377		

Summary & Findings

1. Customer Demographics:

People of all age groups are shopping — from young adults to seniors, buying clothing and beauty products.

2. High-Value Transactions:

Several transactions crossed ₹1000, indicating a few high-spending customers.

3. Sales Trends:

Sales vary across months, helping to identify peak shopping seasons.

4. Customer Insight:

The data reveals the most frequent and highest-spending customers, as well as the most popular product categories.

5. Reporting:

- Sales Summary: Shows total sales, popular categories, and top customers.
- Trend Analysis: Helps identify monthly patterns and plan promotions accordingly.